

HALF/FULL ADDER

Aim:

To realize half/full adder.

- i. Using X-OR and basic gates.
- ii. Using only NAND gates.

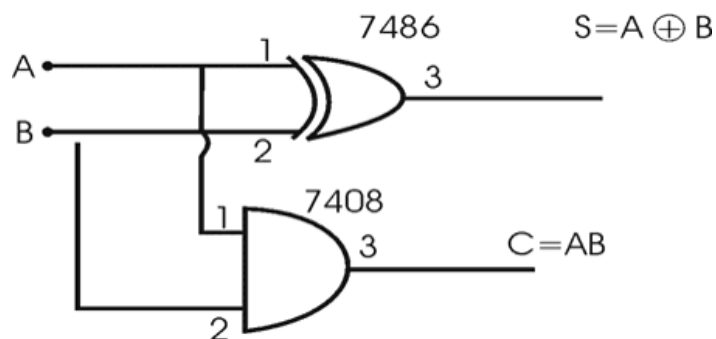
Apparatus Required:

Digital Trainer Kit, IC 7486, IC 7432, IC 7408, IC 7400, and connecting wires

Procedure:

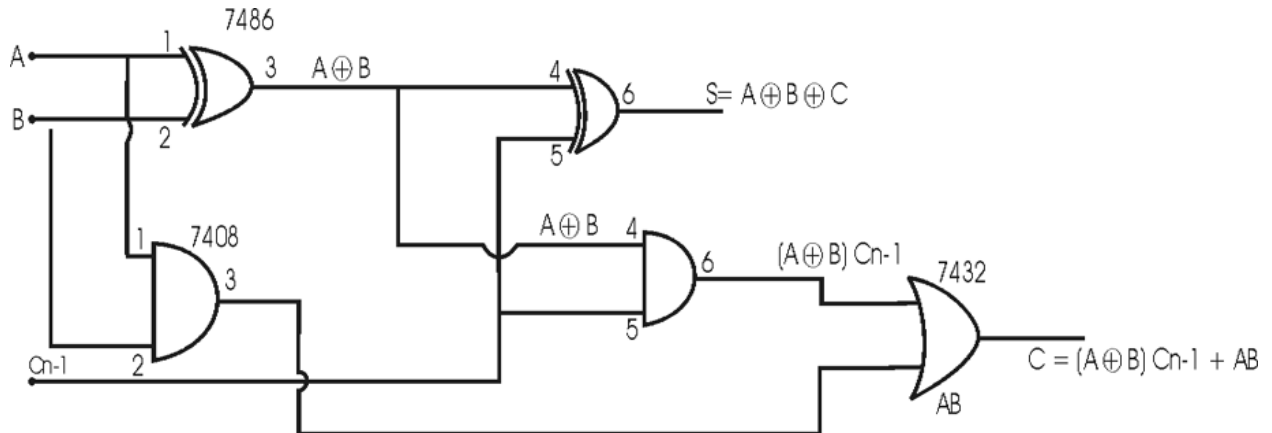
1. Verify the gates.
2. Make the connections as per the circuit diagram.
3. Switch on V_{CC} and apply various combinations of input according to the truth table.
4. Note down the output readings for half/full adder and half/full subtractor sum/difference and the carry/borrow bit for different combinations of inputs.

Half Adder using basic gates:-

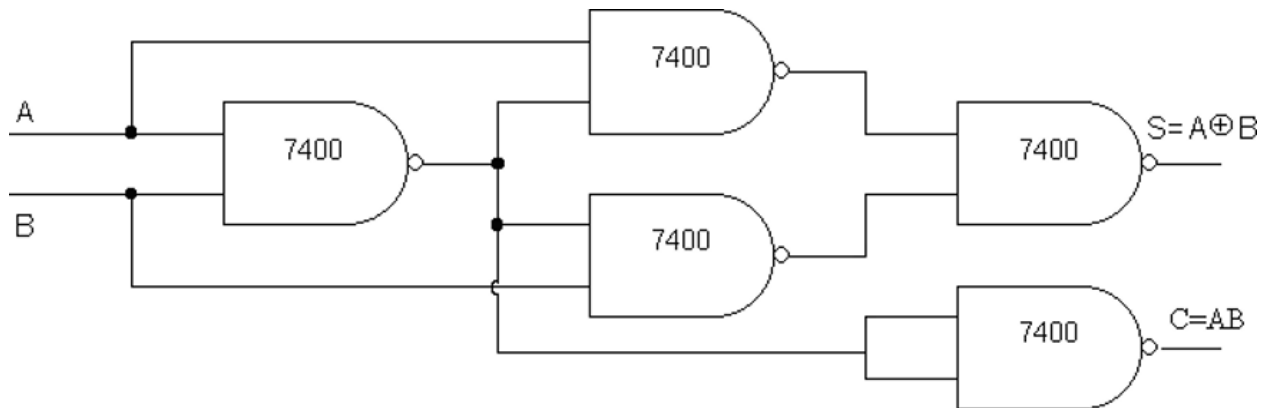


$$S = \bar{A}B + A\bar{B}$$
$$S = A \oplus B$$
$$C = AB$$

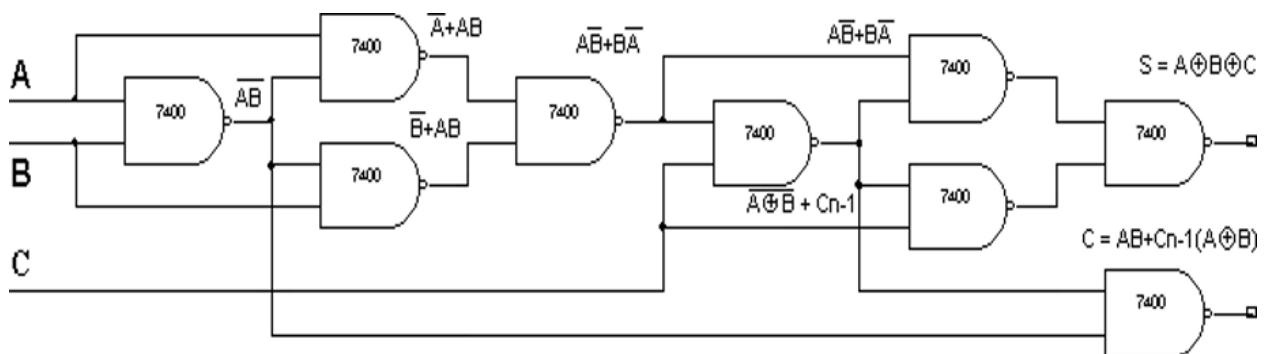
Full Adder using basic gates:-



Half Adder using NAND gates only:



Full Adder using NAND gates only:-



Half Adder					
A	B	S	C	S(V)	C(V)
0	0	0	0	0V	0V
0	1	1	0	5V	0V
1	0	1	0	5V	0V
1	1	0	1	0V	5V

Full Adder						
A	B	C _{n-1}	S	C	S(v)	D(v)
0	0	0	0	0	0V	0V
0	0	1	1	0	5V	0V
0	1	0	1	0	5V	0V
0	1	1	0	1	0V	5V
1	0	0	1	0	5V	0V
1	0	1	0	1	0V	5V
1	1	0	0	1	0V	5V
1	1	1	1	1	5V	5V

CONCLUSION:

Thus, the half Adder , Full Adder circuits were successfully realized and verified using X-OR and basic gates and NAND gates only. The obtained Sum and Carry output were found to be in agreement with respective truth tables.